ARAFAT MAHAMOOD SHAIKH

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PROFESSIONAL SUMMARY

MSc Mechatronics and Robotics student with a Mechanical Engineering background and hands-on experience building and validating electromechanical robotic hardware. Built a tendon-driven wrist exoskeleton end-to-end, covering mechanical assembly, actuator and sensor integration, cable/harness routing, and embedded control, alongside CAE simulation work (ANSYS, SolidWorks, CREO). Works confidently from CAD assemblies and engineering drawings across mechanical, electrical, and firmware disciplines, and documents builds and test results to a professional standard. Passionate about translating robotics research into reliable hardware through rapid prototyping, assembly, testing, and continuous improvement.

EDUCATION

MSc Mechatronics and Robotics — The University of Leeds *2025 – Present*
Relevant modules: Embedded Microprocessor System Design, Medical Electronics & E-Health, Control Systems

B.Tech Mechanical Engineering — BMS Institute of Technology *2021 – 2025*

FEATURED PROJECT — HANDS-ON HARDWARE BUILD

Unilateral Wrist Exoskeleton for Home-Based Stroke Rehabilitation (MSc Dissertation) *Jan 2026 – Present*

- Designed and 3D-printed a modular electromechanical assembly (forearm anchor cuff + dorsal hand-plate end effector) in SolidWorks, including custom actuator mounting brackets and strap hardware for secure donning/doffing.
- Built a tendon-driven parallel mechanism using Dyneema cable routing and multi-motor differential tension across 2 degrees of freedom, integrating and wiring 3x Dynamixel XL-320 smart servomotors onto the forearm assembly.
- Integrated and wired dual Fanspark BMI270 IMUs for real-time orientation tracking and relative kinematics via sensor fusion.
- Developing forward/inverse kinematic solvers (Newton-Raphson) in C++ on an Arduino MKR Zero to drive real-time motor actuation from hand-orientation data.
- Performed subsystem integration, functional testing and iterative validation of the actuator, sensor and control subsystems, identifying and resolving mechanical alignment and control issues during development.
- Authored full build documentation: technical scoping document, Bill of Materials sourced from RS Components/Farnell, and Gantt-based project plan.

ACADEMIC PROJECTS

Voting Machine System — Audio Subsystem (Arm Cortex-A9, DE1-SoC) *Mar 2026 – May 2026*

- Assembled, wired and validated the DE1-SoC audio subsystem, performing hardware integration, debugging and functional testing using bare-metal C, memory-mapped I/O, and hardware-level bitwise register control.
- Integrated and tested the module against the wider system built by a 3-person team, debugging build/link/hardware issues in Arm Development Studio.
- Authored complete system architecture and hardware testing documentation (GitHub README) for handover and viva.

Structural Validation of Ankle Rehabilitation Device *Sept 2025 – Dec 2025*

- Performed Finite Element Analysis (FEA) in ANSYS on a 5-part medical-device assembly, identifying stress concentrations and verifying structural integrity.
- Ran iterative geometric optimisation across design revisions to achieve a Factor of Safety of 1.8, suitable for a medical-grade application.

Experimental Analysis of Ventilation-Based Cooling for BTMS *Nov 2024 – Mar 2025*

- Built and instrumented a physical test rig (replicated battery box + ventilated duct) to measure cooling efficiency across varied battery sizes and thermal conditions.
- Ran controlled experiments across two thermal conditions, recording and analysing results to evaluate ventilation-based cooling performance for automotive battery components.

INTERNSHIP EXPERIENCE

CAD-CAE Intern (Virtual) | Eleation Pvt. Ltd. *Feb 2024 – May 2024*

- Modelled and analysed mechanical components in SolidWorks, CREO, and ANSYS Workbench, preparing simulation-ready geometry via meshing and refinement.
- Conducted static and thermal simulations, including tetrahedral meshing and Von Mises stress analysis, to evaluate component behaviour under load.

TECHNICAL SKILLS

CAD / Mechanical Drawings: SolidWorks, CREO, SolidEdge — reading and producing assembly drawings and electromechanical layouts
Workshop & Assembly: Soldering, standard hand tools, cable/harness assembly, 3D printing, prototype build and rework
Hardware Build & Test: Electromechanical assembly, actuator/sensor wiring and integration, cable/harness routing, functional testing, ANSYS Workbench (FEA)
Embedded & Programming: C, C++, Python, Arduino IDE, bare-metal ARM Cortex-A9, embedded control systems
Documentation: BOM authoring, technical scoping documents, test/result reporting, Gantt project planning, GitHub documentation

CERTIFICATIONS

- AI Tools Workshop — be10x
- Sustainable Energy Technology — IIT Madras (MOOC)
- Basics of Electric Vehicle — ASDC e-Learning Course

LEADERSHIP & EXTRACURRICULARS

- National Cadet Corps (NCC)
- Football Team Member